COMS BC1016

Introduction to Computational Thinking and Data Science

Lecture 14: AB Testing

Spring 2025

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November 9, 2026

Logistics

- Professor office hours are only Monday 4-6 this week
- TA Aarya will be giving a guest lecture on Wednesday!

10	11/9	14 - AB Testing		HW 5 - Testing Hypotheses	
	11/11	15 - Confidence Intervals	Lab 7 - A/B Test	HW 6 - Confidence Intervals	
11	11/16	16 - The Normal Distribution			
	11/18	17 - Sample Means	Lab 8 - Normal Distribution and Variability of Sample Means	HW 7 - Sample Sizes and Confidence Intervals	HW 5 Due
	11/20	-			**Final Project Proposals** Due
12	11/23	Special Topics			HW 6 Due
	11/25	*Holiday*	*No Lab*		

Project Milestones and Deadlines

Next week Friday →

4 weeks away →

~ 5.5 weeks away →

1. **Group Declaration**: Deadline **Friday, Nov 6**
   Please complete the Google Form to indicate your group preference: Final Project Groups. If you do not
   for you, there is a space for you to indicate this. Otherwise, you should indicate who you would like to wo
   name. You may choose whatever name you like as long as everyone inputs the same name. We will be us
2. **Project Proposal**: Due **Friday, Nov 20** at 11:59pm
   Each group will select a final project notebook and dataset to work on for the final project and complete
   Exploratory Data Analysis section. Groups do not need to include sentence explanations of their graphs
   on your exploratory data analysis, you will state the hypothesis (null and alternative) you are planning to
3. **Progress Report**: Due **Monday, Dec 7** at 11:59pm
   For the progress report, groups should complete the exploratory data analysis section and have mostly c
   Specifically, groups are expected to have a clearly stated null and alternative hypothesis, a clearly stated
   implementing the hypothesis test. The code for your hypothesis does NOT need to work for the progress
   whether you have started the hypothesis section. Groups should also begin the prediction section. For w
   section by the time of the progress report, groups should list how they plan on approaching it and what
   groups should share if they are running into any issues with their analysis that they may need assistance
4. **Final Project Report**: Due **Friday, Dec 18** at 11:59pm
   Groups will submit the completed reports along with a completed peer review.

Note: We will require all students to complete a peer review to share how work was distributed among team m
distribution of work will be factored into individual grades on this assignment.

Project Page Updated with Data Sets

- **NYC Restaurant Inspections - Link to starter notebook**

  The New York Department of Health and Mental Hygiene (DOHMH) provides a database of all violations, both confirmed and still being reviewed, from all restaurant and college cafeteria inspections done in the past three years . The Restaurant health data provided was downloaded Oct 2025 and has been separated into three datasets: Grade, Location, and Violation.

- **Seattle Pet Licenses - Link to starter notebook**

  The city of Seattle makes available its database of pet licenses issued from Jan 2017 to Oct 2025 as part of the city's ongoing Open Data Initiative. We have also prepared two additional datasets. The first is the Statistics of Income (SOI) dataset for WA from the 2022 tax year, which features the number of tax returns received by the IRS from each zip code broken out by several income brackets. The second is the Seattle Parks and Recreation Park Addresses.

- **College Scorecard: Colleges, Majors, and Earnings - Link to starter notebook**

  The U.S. Department of Education publishes the College Scorecard to help students compare colleges on cost, debt, graduation rates, and how much graduates go on to earn. The data comes from federal financial aid records, tax records, and reports colleges submit to the government. We have provided 3 datasets from the most recent College Scorecard release (October 2026): information about colleges, information about the majors offered at those colleges, and a table that groups majors into broad fields of study.

- **Duolingo - Link to starter notebook**

  Duolingo is a free language learning app where learners practice words through short lessons. In 2016, Duolingo researchers released a large dataset of learners' practice history along with their paper *A Trainable Spaced Repetition Model for Language Learning* (Settles & Meeder, 2016). The data is available on the Harvard Dataverse. We have provided 3 datasets: a record of practice sessions, a table of the words being practiced, and a small table of facts about each language.

https://www.eysalee.com/courses/f26/bc1016_final.html

If you would like to analyze anything apart from this, please email me for approval

Final Project Proposal

- Choose your project topic and explore the data set!
- For your proposal:
  - Begin the **exploratory data analysis**: 1 quantitative plot, 1 qualitative plot, 1 table made using an aggregate function, and 1 table made using a join
    - You do *not* need sentence explanations for the proposal
  - Propose a **hypothesis** you would like to test (null and alternative)
- Groups of 3 need double the plots, tables, and hypotheses

Final Project Proposal

- Grading is based on completion and overall quality/readability
  - For ease of grading the completion portion, please bold your **null hypothesis** and **alternative hypothesis** and clearly indicate what required component each visualization corresponds to
  - For overall quality and readability, we recommend adding comments into your code and deleting extraneous code
- Submit **as a group** on Gradescope by Friday, Nov 20
  - Upload your .ipynb file and the .csv files

Last Time: Hypothesis Testing and *P*-values

Incomplete Information

- With hypothesis testing, we're trying to choose between two views of the world based on data in a sample
    - It's not always clear whether the data is consistent with one view or another
    - Random samples can sometimes end up in the extremes
        - This is unlikely but possible

Hypothesis Testing

- **Hypothesis Testing:** A statistical test in which we choose between two potential views
  - **Null Hypothesis:** Clearly defined model based on chance. Data is generated randomly and under clearly specified assumptions
    - This is the one we can simulate and test!
  - **Alternative Hypothesis**: The observed data differs from the null hypothesis in some way other than chance
    - Doesn’t say how or why the model isn’t good, just that it isn’t good

Test Statistic

Test statistic: The statistic we choose to simulate to decide between the two hypotheses

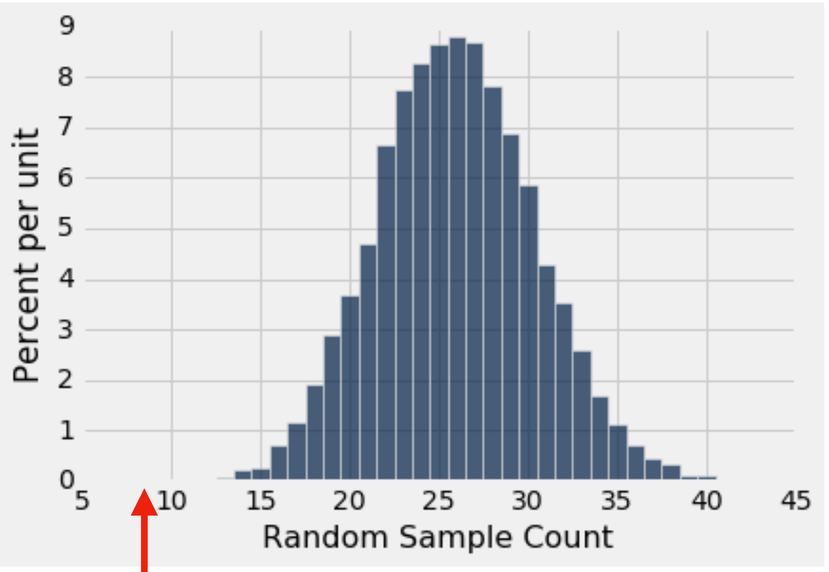
- If we simulate the test statistic under the null hypothesis, we can see what values are likely if the null hypothesis is true

- Where does the test statistic of our observed data fall?

    - If the test statistic of the observed data is highly unlikely under the assumptions of the null hypothesis, we can reject the null hypothesis

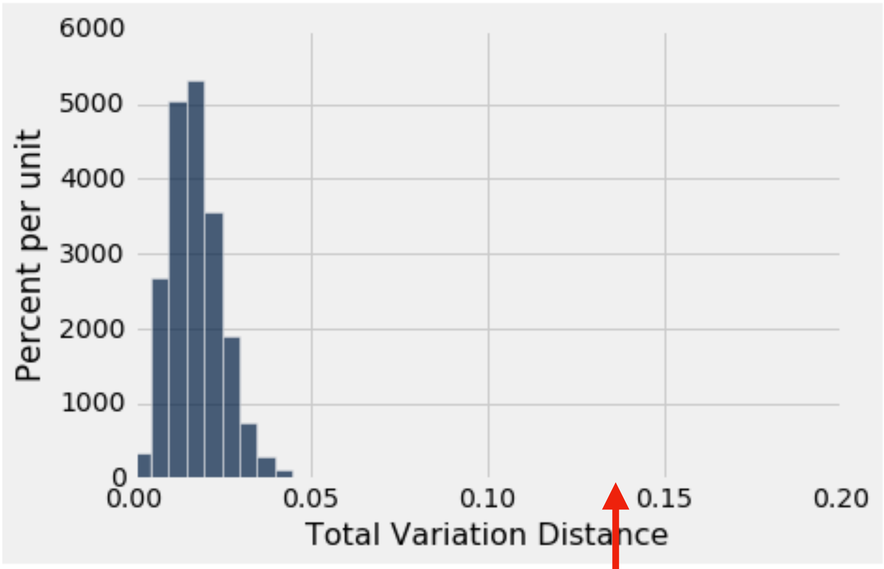
Predictions Under the Null Hypothesis

Swain v Alabama



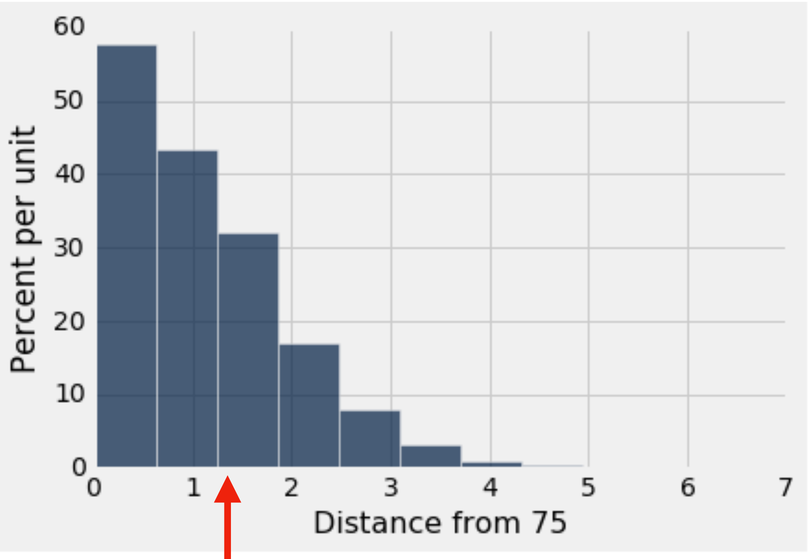
Observed Number (8)

Alameda Jury



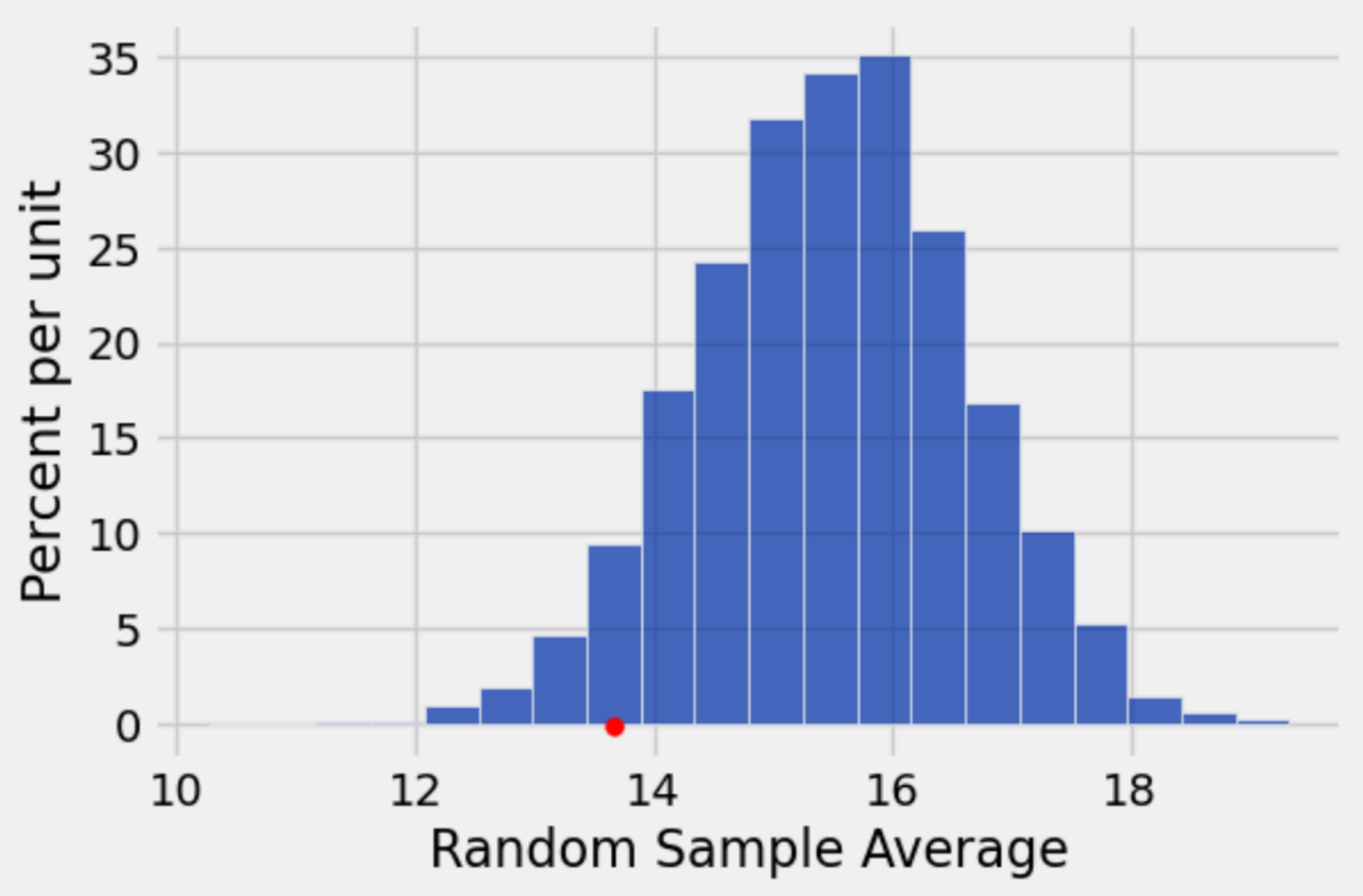
Observed TVD (0.14)

Pea Plants



Observed Distance (1.32)

What if the Conclusion is Unclear?



In a Close Call, Calculate the *P*-value

P-value is a probability that reflects how likely an observed test statistic is under the assumptions of the null hypothesis

- The higher the p-value (near 1): more evidence for the null hypothesis

- The lower the p-value (near 0): more evidence for the alternative hypothesis

Definition of the *P*-value

Formal name: observed significance level

The ***P*-Value** is the chance (probability)

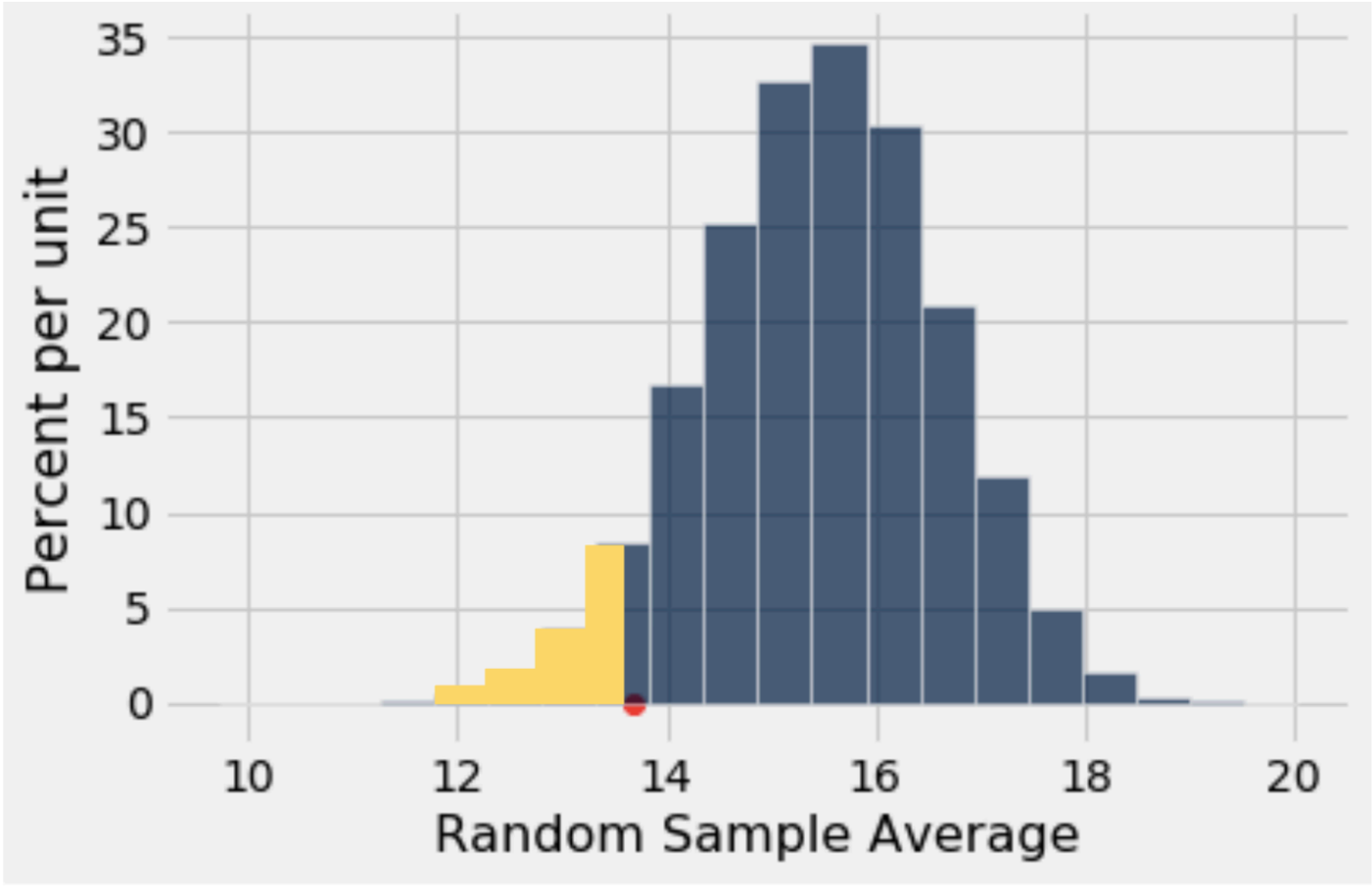
- under the null hypothesis
- that the test statistic
- is equal to the value observed in the data
- or is even further in the direction that supports the alternative

P-Value as an Area

- Empirical distribution of the test statistic under the null hypothesis

- Red dot denotes the observed statistic

- Yellow area denotes the tail probability (p-value)



P-Values and Error Probabilities

A Method for Making Decisions

- Formulate null and alternative hypotheses to decide between
- Observe some data
- Pick a test statistic that can distinguish between the hypotheses
  (and figure out the direction: does a large or small observed test statistic provide evidence for the alternative?)
- Choose a p-value cutoff: What p-values are so small that you'll decide you have enough evidence to reject the null hypothesis?
- Compute the p-value for the observed data and compare it to the p-value cutoff you chose

Discussion Question

There are roughly 3200 students at Barnard.

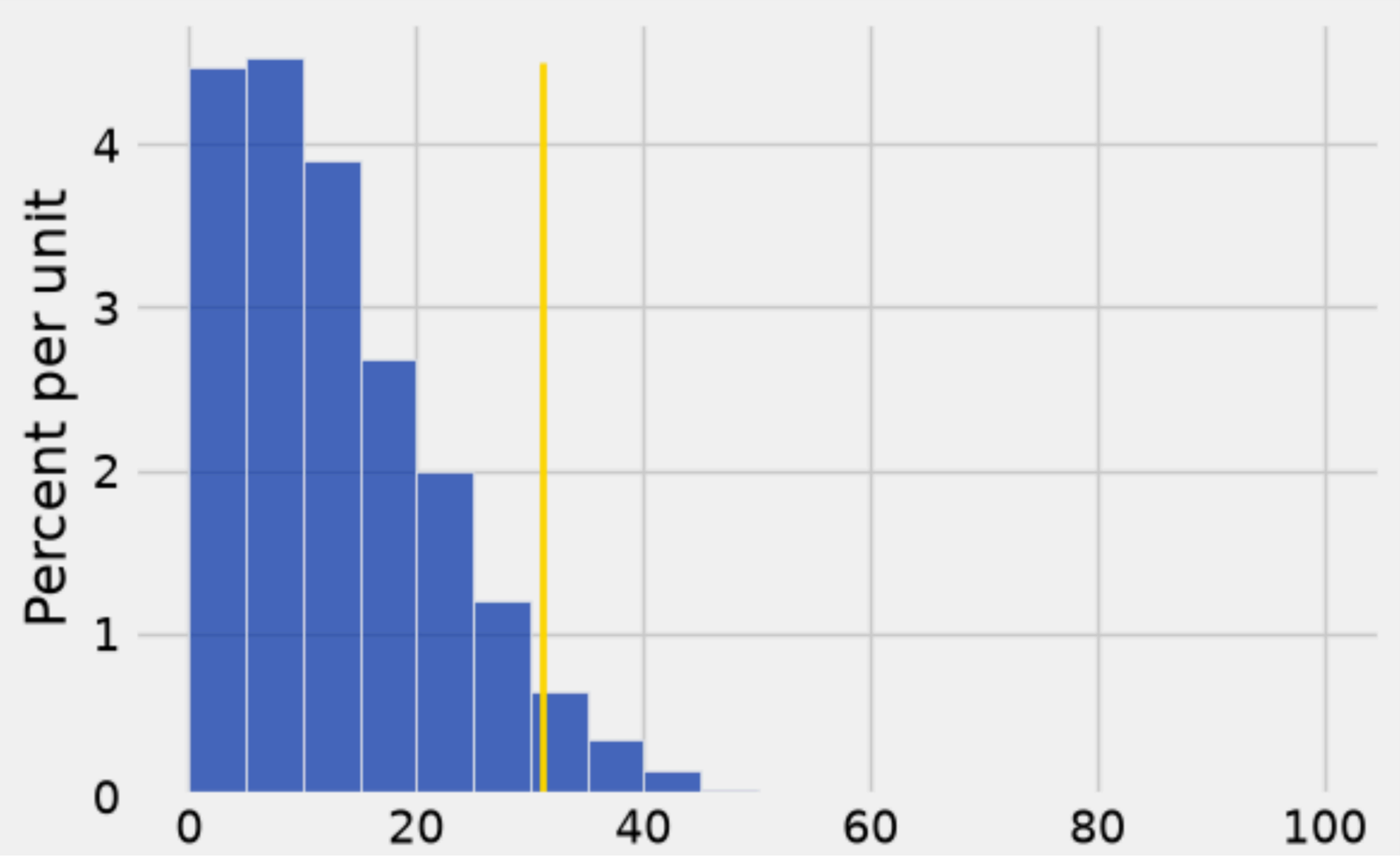
Each student tests:

- Null hypothesis: The coin is fair
- Alternative: The coin is unfair

based on 1000 tosses of a coin, using $\left|\text{number of heads} - 500\right|$ as the statistic, and a 5% cutoff for the p-value

Suppose all 3200 students had fair coins. Will all of them conclude the coin is fair? Roughly how many will conclude the coin is unfair?

Statistic Simulated Under the Null



About 5% of the area is to the right of the gold line

Can the Conclusion be Wrong?

Yes.

	Null is true	Alternative is true
Test favors the null	✓	✗
Test favors the alternative	✗	✓

An Error Probability

- The cutoff for the P-value is an error probability
- If:
  - your cutoff is 5%
  - and the null hypothesis happens to be true,
- then there is about a 5% chance that your test will reject the null hypothesis

P-value cutoff vs P-value

- **P-value cutoff**
  - Does not depend on observed data or simulation
  - Decide on it before seeing results
  - Conventional values at 5% and 1%
  - Probability of hypothesis testing making an error
- **P-value**
  - Depends on the observed data and simulation
  - Probability under the null hypothesis that the test statistic is the observed value or further towards the alternative

A/B Testing

Comparing Two Samples

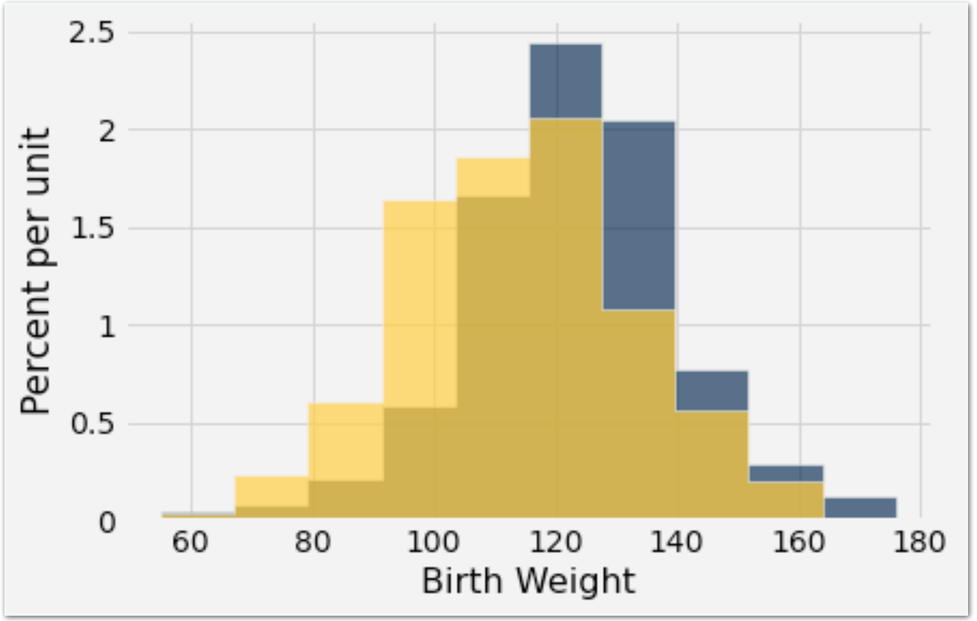
- Compare values of sampled individuals in Group A with values of sampled individuals in Group B
  - Examples:
    - Outcomes in a medical trial (treatment / control group)
    - Outcomes of two different versions of a website
- Question: Do the two sets of values come from the same underlying distribution?
- Answer this question by performing a statistical test is called **A/B testing**

DEMO

The Groups and the Question

Random sample of mothers of newborns.
Compare:

- **Group A**: Birth weights of babies of mothers who smoked during pregnancy
- **Group B**: Birth weights of babies of mothers who didn't smoke during pregnancy



Question: Can the difference in birth weight be due to chance alone?

Hypotheses

Question: Can the difference in birth weight be due to chance alone?

- **Null Hypothesis**: In the population, the distribution of birth weights of babies from both groups are the same.
    - That is, the difference we observe in the sample is due to chance
- **Alternative**: Babies of mothers who smoke weigh less, on average, than babies of non-smokers

Test Statistic

- Group A: non-smokers
- Group B: smokers

- **Test statistic**: Difference between average weights
  - Difference in averages = (Group B average) - (Group A average)
- Negative values of this statistic favor the alternative

How to simulate differences between 2 groups?






Non-Smoker	Non-Smoker	Smoker	...	Smoker
120 oz	113 oz	128 oz		108 oz

Null Hypothesis: the distribution of birth weights of babies from both groups are the same.

Shuffling Labels Under the Null






Smoker

120 oz

Non-Smoker

113 oz

Non-Smoker

128 oz

...

Smoker

108 oz

Null Hypothesis: the distribution of birth weights of babies from both groups are the same.

Shuffling Rows

Random Permutation

- `tbl.sample()`
  - Table with same number of rows as original tbl, picked randomly with replacement
- `tbl.sample(n)`
  - Table of n rows picked randomly with replacement
- `tbl.sample(n, with_replacement = False)`
  - Table of n rows picked randomly without replacement
- `tbl.sample(with_replacement = False)`
  - All rows of tbl, in random order

DEMO

Simulating Under the Null

- If the null hypothesis is true, all rearrangement of labels are equally likely
- **Permutation Test**:
  - Shuffle all group labels
    - Keep the sizes of Group A and Group B same as before, but mix which weights fall into Group A and Group B
  - Find the difference between the averages of two shuffled groups
  - Repeat

Demo: AB Testing Birth Weights

Extra: Causality

Causality

- We saw a relationship between mothers who smoke and birth weights
- Question: Can we say that smoking **causes** lower birth weights?
  - We can't tell!
    - Other factors may contribute to decision to smoke (e.g., income, geography, diet)
  - For observational data (e.g., our Maternal Smoking example) we can claim association but not causation

Randomized Control Experiment

- In data science, the gold standard for determining causation is a randomized control experiment
  - Group A: control group
  - Group B: treatment group
- Participants are **randomly assigned to the groups**
  - Any differences in outcomes between the two groups could be due to chance or the treatment

Next Class

- Today
  - AB Testing
- Wednesday
  - Confidence Intervals